

HARDWARE

TASK 1:

Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

Point of Difference	DDR3	DDR4	DDR5
Speed (Data Rate)	800 – 2133 MT/s	1600 – 3200 MT/s	4800 – 8800+ MT/s
Voltage	1.5 V (1.35 V for DDR3L)	1.2 V	1.1 V
Bandwidth	Low (up to ~17 GB/s)	Medium (up to ~25.6 GB/s)	High (40 GB/s and above)
Typical Use	Older PCs, basic tasks, legacy systems	Modern PCs, gaming, office work	High-performance PCs, gaming, AI, content creation

TASK 2:

Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

Hint: Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.

1. RAM MODULE PRE-INSTALLED
(BEFORE REMOVAL)



RAM module is already installed in the slot and secured with retention clips.

2. RAM MODULE REINSTALLED
(AFTER INSTALLATION)



RAM module is reinstalled and firmly locked with retention clips on both sides.

Steps for RAM Pre-Installation & Re-Installation

1. **Turn off the laptop** and disconnect the charger and battery (if removable).
2. **Open the back panel** using a screwdriver carefully.
3. **Locate the RAM slot** on the motherboard (RAM may already be pre-installed).
4. **For removal:** Gently push the side clips outward and take out the RAM at an angle.
5. **For installation:** Align the RAM notch with the slot correctly.
6. **Insert the RAM** at a slight angle and press it down firmly.
7. **Lock the clips** until they click into place.
8. **Close the back panel** and tighten the screws.
9. **Turn on the laptop** and check if the RAM is detected in system settings.

TASK 3:

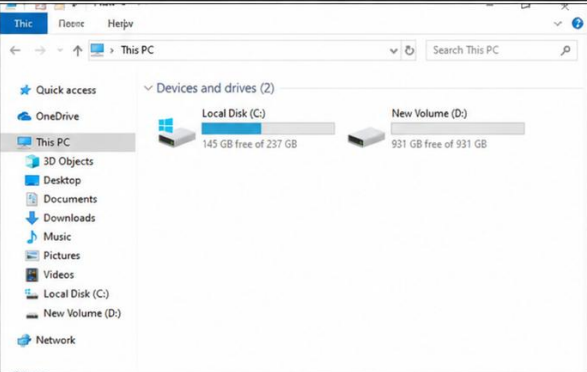
List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Point of Difference	HDD (Hard Disk Drive)	SSD (Solid State Drive)	NVMe (Non-Volatile Memory Express)
Speed	Slow ($\approx$ 80–160 MB/s)	Fast ($\approx$ 200–550 MB/s)	Very Fast ($\approx$ 1500–7000+ MB/s)
Durability	Low (mechanical parts, easily damaged)	High (no moving parts)	Very High (advanced solid-state tech)
Typical Use / App	File storage, movies, backups (e.g., storing videos, documents)	Everyday apps (e.g., Google Chrome, Microsoft Word)	High-performance apps (e.g., Adobe Premiere Pro, modern games)
Voltage / Power Use	High power consumption	Low power consumption	Very low power consumption (efficient)

TASK 4:

Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.

Constraint: Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model.

1. REMOVING M.2 SSD	2. REMOVING NVMe SSD	3. REMOVING SATA SSD
		
Step 1: Unscrew the M.2 SSD	Step 1: Unscrew the NVMe SSD	Step 1: Disconnect SATA data and power cables
		
Step 2: Gently pull out the M.2 SSD	Step 2: Slide out the NVMe SSD	Step 2: Remove the SATA SSD from the bay
4. SUCCESSFUL COMPLETION – REASSEMBLY & VERIFICATION		
		
Reassembled all components and connected cables properly.		System recognizes the storage drives successfully.

The above figure shows the complete process of removing and reinstalling different types of storage devices such as M.2 SSD, NVMe SSD, and SATA SSD. First, the drives are carefully removed by unscrewing or disconnecting the cables, depending on the type of connector used. M.2 and NVMe drives are taken out by loosening a screw and sliding them out at an angle, while the SATA SSD is removed by disconnecting the data and power cables.

After observing the connector types, the drives are properly reinstalled and all cables are securely connected. Finally, the system is powered on, and successful detection of the storage devices confirms that the installation process has been completed correctly.